Anamitra Majumder

Lead Data Scientist and Modelling — Prana Climatech
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Profile

Climate-tech and quantitative systems specialist with strong foundations in statistics, machine learning, and predictive analysis. Designs multi-agent simulators, financial sentiment engines, and climate risk intelligence platforms to forecast markets, collapse dynamics, and environmental resilience.

Education

Indian Institute of Information Technology, Lucknow Ongoing
M.Sc. Data Science — CGPA: 8.2/10
Key Modules: Machine Learning, Deep Learning, Advanced Statistics, Data Structures
Achievements: JAM 2024 (Statistics) AIR 173 TOEFL 108/120

Institute of Mathematics and Applications 2023
B.Sc. Mathematics and Computing — CGPA: 7.10/10

Experience

Lead Data Scientist and Modelling — Prana Climatech Jan 2026 – Present
Leading carbon market intelligence and environmental forecasting pipelines for sustainability platforms.

Data Science Intern — RM Agrico Pvt. Ltd. Jul 2025 – Jan 2026
Built machine-learning pipelines for environmental data modeling.

Volunteer — Climate Resilience Observatory Apr 2025 – Dec 2025
Developed five climate-risk prediction models for resilience planning and lead multiple projects teams.

Teaching Assistant — Probability and Statistics Aug 2024 – Dec 2025
Supported instruction, evaluation, and mentoring of statistics,finance and mathematical methods for data science courses and assisted >120 students per semester.

AI Engineer Intern — INAI World Jul 2025 – Oct 2025
Contributed to 3 trillion parameter LLM system engineering and ML workflow optimization.

Technical Skills

- Programming & Systems: Python, C, Linux, Bash
- Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, Statsmodels
- Big Data: Spark, Hadoop, Hive, Kafka, Dask
- Data & Visualization: NumPy, Pandas, Matplotlib, Plotly, Power BI, Tableau
- MLOps & APIs: Docker, Git, FastAPI, Flask, Streamlit
- Quant & Risk Modeling: Time-series forecasting, systemic risk, Monte Carlo simulation, SHAP interpretability

Soft Skills

- Clear technical communication and scientific documentation
- Collaborative teamwork across interdisciplinary research and engineering teams
- Analytical problem-solving and structured reasoning
- Rapid adaptability in high-velocity technical environments
- High attention to detail in modeling, experimentation, and deployment
- Technical leadership and peer mentoring
- Strong time management across parallel research and production pipelines

Projects

SWANSIM — Black Swan Civilization Simulator — Multi-agent RL simulator modeling macroeconomic collapse and systemic risk with MADDPG agents, scale-free social networks, and a real-time Streamlit dashboard tracking GDP growth, inflation, unemployment, Gini coefficient, firm profit, and social unrest indices.

RedditAlpha — Sentiment-Driven Equity Forecasting — FinBERT-driven Reddit sentiment + deep ARIMA neural forecasting achieving RMSE = 0.00286, MAE = 0.00138, and R^2 = 0.99973 across major Indian equities (per-ticker R^2 up to 0.9981).

ESG-Oracle — Forecasted national ESG trajectories using RNN/LSTM models with R^2 up to 0.951; generated synthetic sustainability data via CVAE.

Global FX Stress Prediction — XGBoost + RNN pipeline forecasting currency stress using geopolitical and macroeconomic indicators with F1 = 0.83 and SHAP interpretability.

Fractal-Based Data Analysis Library — Research library enabling multiscale fractal feature extraction and non-linear complexity analysis of climate and financial time-series.

Dissertation — Poverty, Inequality, and Corruption — Statistical modeling of macro-level socioeconomic interdependencies and inequality dynamics.

Publications

Presented: Statistical Analysis of War and Post-War Economics — 49th OMS Conference, 2022